

STUDENT PERFORMANCE ANALYTICS SYSTEM

Python with AI Internship Report

PROJECT DETAILS	
Project Title	Student Performance Analytics System (BTech Curriculum Edition)
Domain	Data Analysis, Automated Reporting, and CLI Tool Development
Target Audience	Engineering Students, Faculty Evaluators, and Academic Administrators
Student Name	Sachin Yallappa Kumadalikar
Degree & Program	BTech in Computer Science and Engineering
Institution	Bharatratna Indira Gandhi College of Engineering, Solapur (MH)
Affiliation	Dr. Babasaheb Ambedkar Technological University, Lonere, Raigad (MH)

Project Repository

<https://github.com/sachin4749/student-performance-analytics-system-ewb>

1. Title – “Student Performance Analytics System”

2. Objective

The primary objective of the project is to build an automated, robust, and interactive **Student Performance Analytics System** using **Python, Pandas, and NumPy**. The system reduces the need for manual calculations when working with student performance records.

- Ingest raw CSV student records and handle incomplete or missing entries gracefully.
- Calculate total scores and average percentages using vectorized operations.
- Apply standardized academic grading criteria and attendance threshold validation.
- Provide an interactive CLI dashboard for class statistics and performance analysis.
- Provide subject-wise and department-wise analysis.
- Identify top-performing students and support individual student searches.
- Export processed student information into a CSV report.

3. Technologies Used

Python 3.x
Pandas
NumPy
Git & GitHub

4. Dataset Description

The project operates on a structured comma-separated values file named **students.csv**. It contains **25 records** representing **BTech engineering students across Computer Science and Engineering, Information Technology, Electronics & Telecommunication, and Mechanical Engineering**

Column	Description
Student ID	Unique integer identifier ranging from 101 to 125.
Name	Full name of the student.

Column	Description
Department	Engineering branch/stream.
Python	Subject marks out of 100.
SQL	Subject marks out of 100.
C++	Subject marks out of 100.
OOP	Subject marks out of 100.
IOT	Subject marks out of 100.
Attendance	Attendance percentage used in passing criteria validation.

To test data-cleaning capabilities, intentional missing values (NaN) were introduced into specific subject and attendance cells. Missing subject marks are filled using column medians, while missing attendance values are filled using column means.

5. Implementation

5.1 Data Loading and Cleaning

The `load_and_clean_data` function validates that `students.csv` exists, reads the CSV into a Pandas DataFrame, converts subject and attendance columns to numeric values, and imputes missing values so that later calculations do not fail because of NaN entries.

5.2 Vectorized Computation

The `compute_analytics` function converts the subject columns ['Python', 'SQL', 'C++', 'OOP', 'IOT'] into a two-dimensional NumPy array. It then calculates total marks and average percentage using NumPy operations and applies grade and pass/fail rules.

Average	Grade
≥ 85%	A+
≥ 75% and < 85%	A
≥ 65% and < 75%	B+
≥ 55% and < 65%	B
≥ 45% and < 55%	C
< 45%	F

The system applies a dual-condition pass/fail rule: a student passes when **Average ≥ 45%** and **Attendance ≥ 60%**. Otherwise, the student is marked **FAIL**.

5.3 Interactive Terminal Menu

The `display_dashboard` function provides an interactive loop with seven analytical options, allowing users to navigate through the system from the terminal.

6. Key Features

Feature	Implemented Functionality
Overall Class Summary	Total students, class average, highest and lowest scores, pass/fail counts, and grade distribution
Subject-Wise Analysis	Average, maximum, and minimum scores for each individual subject.
Department-Wise Analytics	Comparison of average and peak academic performance across engineering departments.
Top 5 Performing Students	Ranking based on percentage averages.
Student Search	Interactive search by student name or Student ID.
Automated CSV Export	Exports calculated records to processed_student_report.csv.
Robust Error Handling	Graceful handling of invalid user input and missing files.

7. Output

The supplied project output demonstrates the execution of the command-line dashboard and the overall class summary. The following is the recorded dashboard structure:

```
=====
BTECH STUDENT PERFORMANCE ANALYTICS SYSTEM
=====
1. View Overall Class Summary & Statistics
2. View Subject-Wise Performance Breakdown
3. View Department-Wise Analytics
4. View Top 5 Performing Students
5. Search Student by Name or ID
6. Export Processed Report to CSV
7. Exit System
=====
Enter your choice (1-7): 1
```

7.1 Overall Class Summary

Parameter	Recorded Result
Total Students Evaluated	25
Overall Class Average	77.48%
Highest Average Score	95.80%
Lowest Average Score	43.40%
Passed Students	23 (92.0%)
Failed Students	2 (8.0%)

7.2 Grade Distribution

Grade	Number of Students
A+	6
A	9
B+	6
B	3
F	1

8. Final Outcome

The **Student Performance Analytics System** was successfully developed, tested, documented, and published to GitHub. The application demonstrates practical use of Python, Pandas, and NumPy for academic data processing and automated reporting.

The completed system supports CSV data ingestion, missing-value handling, vectorized numerical calculations, grade assignment, attendance-based pass/fail evaluation, class statistics, subject-wise analysis, department-wise analytics, top-student identification, student search, and processed CSV export.

The project repository is:

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